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Drumming Dialogues: Investigating the Fusion of Sabar Drums and Wolof in Senegal

1. Introduction

Drum languages can be defined as communication systems that transpose selected acoustic aspects of spoken languages into rhythmic patterns, conveying a diverse range of messages. They can be divided into two types: those based on tonal languages, which transpose tone melodies, and those based on non-tonal languages; however, those based on tonal languages are the most well-documented. This paper examines some linguistic aspects of the second type of drum languages, namely, drum language from Senegal. Senegalese drummers have a tradition of playing drums in correlation to speech in Wolof, a Niger-Congo language predominantly spoken in Senegal, as well as in the Gambia and the neighboring regions.

Drummed speech has primarily been documented within tonal languages. For instance, among the Yoruba people of Nigeria, drums mimic the Yoruba language (Euba 1990, Villepastour 2010), transposing melodies of a spoken language. The correlation between Wolof and Sabar drumming differs due to the non-tonal nature of Wolof (understanding how specific sounds in spoken language are represented and Winter 2014, Tang 2007).

The present research aims to uncover the underlying regularities between Wolof linguistic units and Sabar drum rhythms. This study analyzes the linguistic aspects of the Sabar drum language through the lens of **Phonological Theory**, that provides a robust framework for understanding how specific sounds in spoken language are represented and transformed in drum language. I hypothesize that the Sabar drum language encodes auditory features of spoken language by transposing characteristics of speech sounds into rhythmic patterns.

Sabar drummers belong to a social class of griots (Wolof: géwël) (Hale 1998, Tang 2007), and their most common drum is a single-headed drum known as sabar. The griot drummers do not always play drums in correlation to speech, the rhythms they play sometimes. However, in this paper I focus on linguistically meaningful sabar rhythms and refer to them as Sabar. In Senegal, sabar drumming is prevalent across various events, including sporting events, life-cycle ceremonies, and political gatherings. While the sabar drums are now mostly used for entertainment purposes rather than as a direct means of conveying linguistic messages, they remain closely linked to linguistic expressions (Winter 2014).

This paper aims to deepen our understanding of the correlation between the Sabar drum language and Wolof, specifically examining the relationship between drum strokes and the [ATR] feature of Wolof vowels. The findings of this study indicate the presence of consistent patterns between Sabar drum strokes and Wolof syllables. The syllable has gained increasing significance in phonological theory, with all major approaches recognizing it as a fundamental unit of analysis, from the early Prague School to modern generative approaches (Goldsmith 2011, Blevins 1995).

First, I provide an overview of the basic phonemic units of Sabar (2.1) and previous attempts to establish the correlation between Wolof and Sabar (2.2). I then delve into the description of the Advanced Tongue Root ([ATR]) feature of the Wolof language, which is significant for the present study, in Section 3. Details regarding the materials and methods used in the research are described in Section 4, followed by a discussion of the results in Section 5. The appendix provides some examples of the data used in the research.

2. **Overview of Sabar drumming tradition in Senegal**

2.1. Basic phonemic units of Sabar

Playing Sabar involves at least nine different drum strokes, which can be seen as the basic units of the genre. These strokes appear in longer sabar rhythmic units which can be correlated with spoken utterances in Wolof. Sabar sounds are produced by one hand (palm or fingers) and one stick. Both the stick and the hand are used for beating rhythms and for damping the sound (Winter 2014). Sabar basic units, which I call Sabar phonemes, are produced by different combinations of applying the hand and the stick onto certain parts of the sabar head. Sabar phonemes can be divided into the following three classes:

• Finger strokes: produced by one hand, which may bounce or stop on the skin.

• Stick strokes: produced by the stick, where the hand may be used for attenuating the sound.

• Hand + stick strokes: sequences of hand strokes and stick strokes; these sequences are perceived as minimal rhythmic units.

Each of the Sabar phonemes has an oral code that griots use when referring to rhythms. The main Sabar phonemes and their oral codes are given in Table 1. The table is taken from (Ros 2021).

	Mnemonic	**Description**
Hand strokes		
gin	gi, bin	Bass sound; fingers strike the edge of the drum or the middle of the skin.
pin		Fingers strike the edge of the drum

pax	pa, ba, bax, *mbar, mbax*	The full hand (the palm and the fingers) strikes the whole skin.
Stick strokes		
tan	ta, sa, san, dan, ja	The stick strikes the centre of the drum and bounces off.
tac	tas, *tach*	The stick strikes the centre of the drum and bounces off, while the hand damps the edge of the drum.
ce	ña, ca, cek, cex, te	The stick strikes the drum and is left there.
Hand+stick strokes		
rwan	Rwa	pin tan or pax tan
rwe	Rwex	pax ce
drin		tan gin

TABLE 1. Sabar phonemes from Ros (2021, p. 2)

The code for any given phoneme refers to the way the sound is perceived, not to the precise technique of its production, which may vary between different types of sabar drums (Tang 2007). The codes can be called slightly differently, depending on personal choice of a drummer, speed of playing or the drum that is being used.

2.2. Previous research on the Sabar drum language

A previous attempt to map Sabar to Wolof was taken in Ros (2021). That research is based upon the same dataset, collected in Senegal, consisting of 402 rhythmic phrases, followed by Wolof translations; it will be described in more detail in the next section. To analyze the collected dataset, two hypotheses were explored: the word-level hypothesis and the syllable-level hypothesis. The word-level hypothesis posits that each Wolof word is associated with a specific Sabar stroke or combination of strokes, while the syllable-level suggests that syllables in Wolof correspond to specific drum strokes, where the nature of the correspondence depends on the phonetic or phonological properties of a vowel in a syllable. The dataset was analysed using statistical methods to identify regularities between Wolof units and Sabar strokes. No validation for word-level hypothesis was found, however, results revealed significant associations between Sabar strokes and Wolof syllables, that depends on the phonological properties of a vowel in a syllable, namely vowel position and vowel openness. In Table 2 the results of Ros (2021) are presented.

Strokes' preferences for vowel position	
ce	mid-open/mid-closed
gin	Closed
pax	open
rwan	open
rwe	mid-open/mid-closed
tac	mid-open/mid-closed
tan	open
Strokes' preferences for vowel openness	
ce	front
gin	back (and front)
pax	central
rwan	central
tac	front
tan	central

TABLE 2. Mapping between the Wolof vowels and Sabar strokes in Ros (2021, p. 5).

These findings laid the groundwork for the current research, which aims to further explore and understand the linguistic properties of the Sabar drum language.

3. **[ATR] in Wolof**

The previous analysis has revealed that different drum strokes are often associated with various types of vowels, and the critical factors are vowel openness and vowel position. These parameters are the basic parameters for representing vowel systems and for this reason they are reflected in the Sabar phonology.

As previously mentioned, unlike other drum languages that mimic the pitch levels of tonal languages, Wolof and Sabar lack tonality, which requires a different analytical approach. To expand our research, I turned to studies on whistled languages, where a greater number of non-tonal languages have been described. Works on whistled languages show that they encode auditory features of spoken languages by transposing key components of speech sounds, namely, vowels. For example, the best-known whistle language – Silbo Gomero, the Spanish-based whistled language of La Gomera – represents the basic phonemic units of spoken Spanish, namely by transposing F2 contours (Rialland 2005).

Pevious research did not analyze phonological feature of Wolof vowels as [ATR], or advanced tongue root, which has a first formant as the main acoustic correlate (Beltzung: 238; Halle & Stevens 1969: 38). The present analysis opted to incorporate this feature, hypothesizing that it may also be reflected in drum phonology, since it plays a significant role in the organization of Wolof vowels (Kaji 1997).

[ATR] refers to the position of the tongue root when a vowel is produced, which affects the vowel's quality. [+ATR] vowels are the vowels with advanced tongue root position, while the [-ATR] vowels are associated with the retracted tongue-root position. [-ATR] vowels are often described as open, lax, light, hard, creaky, narrow or dull. [+ATR] vowels are described as close, tense, heavy, hollow, breathy, wide or bright (Tucker 1964, Lindau 1978, Rottland 1980, Beltzung and Clements 2015, among others). Wolof has five [+ATR] vowels: /i, e, u, o, ë/ and three [-ATR] vowels: /ɔ, ɛ, a/.

The vowel inventory of Wolof is given below in Table 2 (Dye 2015: 20-22). The [ATR] feature has a distinguishing function in Wolof (cf. *reer-oon* (was lost) vs *rɛɛr-ɔɔn* (had dinner)). Wolof also exhibits a pattern of vowel harmony which involves categorizing all vowels in the language into two harmonic groups with regard to the feature [ATR].

The vowel inventory of Wolof is given below, in Table 3 (Dye: 20-22)

	[+ATR]	[-ATR]	[+ATR]	[-ATR]
High	i / ii		u / uu	
Mid	e / ee	ɛ / ɛɛ	o / oo	ɔ / ɔɔ
Low				a / aa

TABLE 3. The vowel inventory of Wolof (Dye: 20-22).

This paper aims to examine possible correlations between the [ATR] feature in Wolof vowels and Sabar strokes. According to the guiding hypothesis, some strokes are more commonly used to encode a set of vowels with [+ATR] feature, while the other strokes are more commonly associated with the set of vowels with [-ATR] feature.

4. **Materials and Methods**

4.1. Materials

This work relies on research materials collected during previous expeditions to Senegal. They include traditional *bakks* — texts and phrases that are associated to sabar rhythms, which are known to many griots, and improvised texts. Improvised texts are spontaneous compositions that griots create on the spot during performances, they are often tailored to the immediate context, such as current events, the audience, or specific individuals present. The recordings were made in live sessions with the drummers. The work was conducted in the years 2018–2019 in Campement Nguekhohk, Senegal. All the recordings were made with griots of the same family, where 2–3 drummers were present in each session. The drummers were asked to come up with a traditional bakk or improvisation in Wolof, play the corresponding rhythm and utter the rhythm's oral codes. Recordings start with the phrase in Wolof, which is followed by the corresponding rhythm. For each recording there was a transcription of both the text and the rhythm. To transcribe the rhythm, the sabar stroke coding system was used, without any detailed annotation of temporal relations between strokes or their acoustic properties.

A total of 402 recordings were made. Of them, six recordings were excluded because the transcription of the sabar rhythm was missing. Of the remaining 396 pieces, 35 are bàkks and 361 are improvised texts. The average number of words and syllables per piece in the spoken language is 11 and 15, respectively, meaning that most of the words are monosyllabic. The average number of strokes per piece is 15. To gain more insight into Sabar practices, fifty of the Wolof pieces were translated into English by a Wolof speaker.

The orthography in the original dataset did not indicate any difference in [+ATR] and [-ATR] mid vowels: both [o] and [ɔ] were written down as o and both [e] and [ɛ] were written down as e. Despite the acoustic differences between these vowels, I was unable to discern them audibly. Therefore, with the assistance of a native Wolof speaker and consultation with a dictionary (Diouf 2003), the dataset had to be revised. Despite these efforts, some words remained problematic, and certain phrases had to be omitted. Consequently, the final dataset consisted of 295 rhythmic phrases.

Example (1) illustrates an improvised text. The text in Wolof is followed by the codes of the corresponding drum strokes.

(1) Mbëg-geel (Wolof)
gin gin (Sabar)

(love)

Kaay	nu	bëg-gën-te	tey	wec-coo	xa-laat (Wolof)
rwan	gin	tan tan gin	rwan	rwan gin	pax tan (Sabar)

(let's love each other today and exchange thoughts)

Mooy	jàm-mi	bo-room	waaw	waaw		(Wolof)
gin	gin gin	gin gin		pax	gin	(Sabar)

(it's the peace of God yes yes)

4.2. Methods of analysis

According to the syllable-level hypothesis, syllables in Wolof have specific drum strokes associated to them, where the nature of the correspondence depends on the phonetic or phonological properties of the vowel in a syllable. In (Ros 2021) the analysis has shown that different drum strokes are more commonly associated with different types of vowels (front, central or back; open, mid-open/mid-closed or closed vowels).

This paper examines whether [ATR] feature plays a role. Therefore, all the syllables were divided into two groups: syllables with a [+ATR] vowel and syllables with a [-ATR] vowel. From the total of 4047 analysed Wolof syllables, 1655 (41%) contained [+ATR] vowels and 2419 (59%) [-ATR] vowels.

To test the syllable-level hypothesis, the data were presented in a table where in each row there was a specific stroke, an associated syllable and the +/- ATR property of the vowel that corresponds to this stroke. Table 4 shows an example of restrucured data, where Wolof_ATR stands for the ATR property of the vowel in the Wolof syllable associated with the given stroke. This data was analysed using SPSS software, specific statistics follows.

Sabar	**Wolof**	**Wolof_ATR**
rwan	A	0 [-ATR]
gin	I	1 [+ATR]
tan	A	0 [-ATR]
tan	U	1 [+ATR]

TABLE 4. Example set of a restructured data.

4.3. Results

A chi-square test of independence was conducted between the stroke type and the ATR feature of the vowel in the corresponding syllable. All the expected cell frequencies were greater than five. There was a statistically significant association between the stroke type and the length of the vowel in the corresponding syllable, $\chi2(7) = 939$, $p < 0.0005$. The association was strong, Cramer's V = 0.48 (Cohen 1988).

A cross-tabulation of stroke types and vowel lengths is presented in Table 5. Adjusted residuals appear in parentheses below the observed frequencies. A residual is the difference between the expected frequency and the observed frequency. The residuals are standardized so that they have an approximately standard normal distribution with the approximation improving at larger sample sizes. The adjusted standardized residual higher than 3, mark the cells that deviate significantly from independence (Agresti 2007). Absolute adjusted standardised residuals greater than three are highlighted in bold.

	ATR-	ATR+
Ce	93 (1,1)	53 (-1,1)
Drin	10 (-1,1)	11 (1,1)
Gin	**435** **(-30,1)**	**1065** **(30,1)**
Pax	**224** **(5,4)**	**78** **(-5,4)**
Rwan	**440** **(10,4)**	**113** **(-10,4)**
Rwe	12 (2)	2 (-2)
Tac	**153** **(3,2)**	**67** **(-3,2)**
Tan	**1052** **(18,4)**	**266** **(-18,4)**

TABLE 5. Cross-tabulation of stroke types and the ATR feature of the vowels.

The statistical analysis therefore shows that some regularities between sabar strokes and Wolof syllables exist, namely the following (only the results with absolute adjusted standardized residuals greater than three are presented):

Gin	ATR+
Pax	ATR-
Rwan	ATR-
Tac	ATR-
Tan	ATR-

TABLE 6. Regularities between sabar strokes and the ATR feature.

5. **Discussion**

Sabar, as a non-tonal language, presents an intriguing subject of study because it operates differently from tonal drum languages that simply mimic the tones of spoken languages. This research represents one of the few attempts to unravel the workings of such languages. Building upon Ros' (2021) work, this paper explores correlations between Wolof vowels and Sabar drum strokes. The findings suggest that Sabar griots use the correlation between the Wolof vowels and Sabar strokes during improvisation, mimicking not the tone but the phonological properties of vowels, suggesting potential similarities with tonal drum languages or whistle languages.

However, these findings should be considered preliminary due to limitations of this work. The correspondence between the Wolof syllables and the sabar strokes, observed in this study, merely outline a possible situation rather than establish definitive rules, and different datasets might yield varying results. For instance, Winter (2011) provides alternative perspectives that differ significantly from the results presented here. Winter states that Sabar involves language-like grammar rules, different from the grammar of Wolof, and that the rhythms do not mimic speech. This highlights the importance of further investigation with diverse data sources to either substantiate or refine these initial observations.

Interestingly, it seems that the drummers themselves might not consciously be aware of these linguistic rules, as they likely internalize them through cultural immersion. Nonetheless, there are strong indications that Sabar phonology does indeed reflect the most significant parameters for representing Wolof vowel systems. However, it's important to acknowledge the limitations of this study. The dataset, collected from just one family, is inherently limited. Future research should aim to address this limitation by incorporating a more diverse sample.

This research significantly contributes to our understanding of the intricate relationship between drum language and the phonological features of spoken language, illuminating how linguistic elements are encoded within drumming traditions and underscoring the cultural significance and complexity of such communication systems. Looking ahead, future research could focus on conducting more extensive studies on other non-tonal drum languages to deepen our understanding of diverse linguistic and musical practices across different cultural contexts.

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Appendix

(1) Bakk about child

Yaay	boy	baay	boy	(Wolof)
rwan	gin	rwan	gin	(Sabar)

mammy daddy

Sama	doom	loo	wax	(Wolof)
tan rwan	gin	rwan	rwan	(Sabar)

my child what did you say

(2) Bakk about a call

Wuyulki	lay	woo		(Wolof)
gin gin	gin	tan	rwan	(Sabar)

respond to the person who calls you

Ki	la	woo	(Wolof)
gin	tan	rwan	(Sabar)

who calls you

War	na	la	woo		(Wolof)
rwan	tan	tan	rwan		(Sabar)

should call you

Wuyulki	lay	woo		(Wolof)
gin gin	gin	tan	rwan	(Sabar)

respond to the person who calls you